

# Immersive Fire Training with Live Expert Support and Embodied Interaction with Real Extintor

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Figure 1: Overview of the Immersive Fire Training Application. Above) Left) View of the Remote Expert which includes the 3D position of the Avatar within the 3D environment, and a 2D window representing their egocentric video and Right) Remote Expert wearing the HMD. Below) Left) View of the trainee seeing-through their own body and real extinguisher and Right) detail of the real extinguisher with controller and HMD with stereo camera.

## ABSTRACT

In this paper, we developed an immersive training environment for mastering the use of a fire extinguisher, with the possibility of receiving live feedback from a remote expert. To facilitate communication and knowledge sharing between trainees and remote experts, we provide bidirectional audio and event synchronization. As additional novel features, we allow the user to see and interact with their own body and real fire extinguisher using deep learning and color-based semantic segmentation. We also allow the expert to monitor the trainee by rendering a 2D video of their egocentric view. Preliminary results show the benefit of receiving live feedback and the potential of using a real fire extinguisher.

**Index Terms:** fire training, immersive environment

## 1 INTRODUCTION

Fire is a challenging element to manage, and unfortunately, a significant portion of the general population lacks even basic training to address potential fire-related accidents. These accidents often result in substantial economic losses and can seriously impact individuals' lives and health. Therefore, it is crucial to promote training and awareness on this matter, both by public institutions (targeting citizens [1]) and private companies (for their employees).

Due to the risks associated with simulating fires in real-world conditions, virtual reality (VR) emerges as a highly useful and convenient tool [7], which has attracted the interest of academia, due to all the technological challenges involves, but also enterprise that envision a business opportunity [2, 3]. VR enables the recreation

of realistic scenarios without the hazards of traditional in-person training (cost, environment, logistic), while offering several advantages (learning enhancement [6], scalability) over more traditional methods. Previous fire training VR works were primarily developed using pure virtual fire extinguishers, but more recent studies have started incorporating real fire extinguishers equipped with tracking technology [5].

In this paper, we develop a prototype of a fire training immersive system designed for two users: a trainee, and a remote expert that by wearing Head Mounted Display (HMD) experience the same virtual environment. The following features enhance the immersive training application: *i*) bidirectional audio communication between trainee and remote expert, *ii*) possibility of the trainee to see their own real body and the real fire extinguisher, and *iii*) possibility of the remote expert to follow the trainee activities by two visual information sources: the 3D position of the trainee, and a 2D window rendered in the 3D environment showcasing the first-person view of the trainee within the environment.

## 2 VISIBILITY, COMMUNICATION AND INTERACTION FEATURES

### 2.1 Visibility

We develop a multi-user Unity project that allows both trainee and remote experts to share the same 3D virtual environment. For the purpose of fire training, we recreated an office where a small fire of type B emerges from a trash bin.

From a first point of view, the trainee can see their own body rendered in the 3D environment, using a video-based self-avatar [4]. The egocentric video captured from the trainee HMD is sent to a machine learning server allocated in the trainee's computer, where a deep learning-based semantic segmentation inference is performed at frame level to get the trainee body. The round trip time including inference of a  $640 \times 480$  image is about 30-40 ms. A

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color-based segmentation is performed directly within Unity to get the real extinguisher from the egocentric video. Both deep learning and color-based segmentation masks are then joined in a Unity shader that renders both the body and the extinguisher on top of the 3D environment (see Fig.1 below).

From a third point of view, the trainee is rendered as an upper body avatar (see Fig.1 above) with a virtual extinguisher<sup>1</sup> and are positioned in the 3D environment thanks to the tracking information provided by the Meta HMD (trainee) and controller (extinguisher).

## 2.2 Communication

We also developed an event, audio, and vision communication kit that enables trainees and remote experts to be aware of each other and give proper feedback. For the bidirectional audio communication, we integrate a plugin for Mumble Audio Library that works in a client-server mode and allows audio communication with a maximum delay of 800 ms.

In addition to the trainee's third-person view, we incorporated the possibility of seeing a first-person view of the trainee's actions. A proprietary library based on GStreamer was developed to capture and send video streams at the network level and in real time. This egocentric video is rendered as a floating window within the inverse environment of the remote user. A detailed of this floating window can be seen in Fig.1 above.

## 2.3 Interaction

For event synchronization, we use NET Code library, due to its peer-to-peer protocol and its native Unity integrations. More concretely, the position and rotation of both the trainee and the extinguisher are the ones that are sent from the trainee to the expert session. Synchronization time is about 300 ms.

To track the fire extinguisher position within the 3D environment one of the Meta Controllers was attached to the horn/exit of the extinguisher, with the help of some zip ties. The use of a Bluetooth joystick transmitter receiver allows the Unity application to know when the extinguisher trigger is activated, to synchronize the extinguisher's foam release animation. The latency between the joystick triggering and the virtual animation in Unity is about 330 ms.

## 3 SYSTEM: SOFTWARE AND HARDWARE REQUIREMENTS

The immersive training application was built using two GPU-powered computers, both running Unity v2020.3.42f. Pycharm is used for running the deep learning semantic segmentation network with Python 3.8 version and Cuda 10. Communication between Unity and Pycharm was done using ZeroMQ TCP-based communication protocol. Both Mumble and Gstreamer are used under an open license.

As for hardware equipment, two Meta Quest 2 are used, one for the trainee, and one for the remote expert, both working in link mode tethered to the GPU computers. Additionally, to enable video-passthrough, a stereo camera is placed in front of the trainee HMD with the help of a 3D-printed case. This allows the possibility to see-through their real body and real fire extinguisher.

A carbon dioxide (CO<sub>2</sub>) fire extinguisher was used. Those areas of the extinguisher that were not red were painted to ease the color-based segmentation process. A Bluetooth joystick was glued to the extinguisher handle to track the trigger of the extinguisher. A small spring piece was helpful to connect the horn with the joystick button.

## 4 PRELIMINARY RESULTS

We conducted a preliminary user evaluation with a group of 16 users (7 women, 9 men; 43.41 average age, standard deviation

<sup>1</sup>both avatar and virtual extinguisher are taken from [www.turbosquid.com](http://www.turbosquid.com)

11.60). Among them 69% were having more than 10 times experience with VR, 13% reports 10-30 times experience, and 19% reports more than 30 times. After trying the demo as a trainee we ask the cohort the following selection of questions taken from different standard questionnaires (items 1,2,3 and 4) and ad-hoc ones (5, 6, 7, and 8).

- **1 Presence** - *I felt that I was inside the office (Yes, Sometimes, No).* 75% answered yes, and 25% answered sometimes.
- **2 Presence** - *Could you concentrate on the fire extinguisher task (Yes, Sometimes, No).* 75% answered yes, and 25% answered sometimes.
- **3 Embodiment** - *Was the body that you could see through the HMD your own body).* 93.8% answered yes, and 6.3% answered sometimes.
- **4 Co-presence** - *I felt that the expert was with me (Yes, Sometimes, No).* 81.3% answered yes, and 18.8% answered sometimes.
- **5 Communication** - *I felt that the expert was with me (Yes, Sometimes, No).* 81.3% answered yes, and 18.8% answered sometimes.
- **6 Communication** - *I could understand clearly the instructions provided by the expert (Excellent, good, poor).* 75% answered excellent, and 12.5% reported good, and 12.5% reported poor.
- **7 NPS** - *What potential do you see to the VR as a practical training tool (from 1 to 10).* 50% turns out to be promoters, rating with a 10, and 50% rating with an 8 (passive).
- **8 General Overview.** *Please rate how have you liked this experience (1 to 5).* The average response was about 4.25

## 5 CONCLUSION

In this work, we have developed an immersive fire training application designed to equip trainees with essential fire-handling procedures with the possibility of seeing your own body and using as real extinguisher. The application incorporates the support of a remote expert, who can monitor and guide the trainee through shared access to the virtual environment and a first-person view of the trainee's actions. For future development, we aim to conduct a more extensive user study and expand the application by introducing a wider variety of scenarios and fire extinguisher types.

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